1 Absolute values and discrete valuations

1.1 Introduction

At its core, number theory is the study of the integer ring $\mathbb{Z}$. By the fundamental theorem of arithmetic, every element of $\mathbb{Z}$ can be written uniquely as a product of primes (up to multiplication by a unit ± 1), so it is natural to focus on the prime elements of $\mathbb{Z}$. If p is a prime, the ideal $(p) := p\mathbb{Z}$ it generates is a maximal ideal ($\mathbb{Z}$ has Krull dimension one), and the residue field $\mathbb{Z}/p\mathbb{Z}$ is the finite field $\mathbb{F}_p$ with p elements (unique up to isomorphism). The fraction field of $\mathbb{Z}$ is the field $\mathbb{Q}$ of rational numbers. The field $\mathbb{Q}$ and the finite fields $\mathbb{F}_p$ together make up the prime fields: every field k contains exactly one of them, according to its characteristic: k has characteristic zero if and only if it contains $\mathbb{Q}$, and k has characteristic p if and only if k contains $\mathbb{F}_p$.

The structure of the ring $\mathbb{Z}$ and the distribution of its primes are both intimately related to properties of the Riemann zeta function

$$\zeta(s) = \sum n^{-s} = \prod_p (1 - p^{-s})^{-1}.$$

As a function of the complex variable s, the Riemann zeta function is holomorphic and nonvanishing on $\mathrm{Re}(s) > 1$ and admits an analytic continuation to the entire complex plane. It has a simple pole at $s = 1$, which implies that there are infinitely many primes (otherwise the product over primes on the RHS would be finite and converge). The distribution of its zeros in the *critical strip* $0 < s < 1$ is directly related to the distribution of primes (via the *explicit formula*, which we will see later in the course). As you are probably aware, Riemann famously conjectured more than 150 years ago that the zeros of $\zeta(s)$ in the critical strip all lie on the *critical line* $\mathrm{Re}(s) = \frac{1}{2}$; this conjecture remains open.

One can also consider finite extensions of $\mathbb{Q}$, such as the field $\mathbb{Q}(i) := \mathbb{Q}[x]/(x^2+1)$. These are called *number fields*, and each can be constructed as the quotient of the polynomial ring $\mathbb{Q}[x]$ by one of its maximal ideals; the ring $\mathbb{Q}[x]$ is a principal ideal domain and its maximal ideals can all be written as (f) for some monic irreducible $f \in \mathbb{Z}[x]$. Associated to each number field K is a zeta function $\zeta_K(s)$, and each of these has an associated conjecture regarding the location of its zeros (these conjectures all remain open).

Number fields are one of two types of *global fields* that we will spend much of the course studying; the others are *global function fields*. Let $\mathbb{F}_q$ denote the field with q elements, where q is any prime power. The polynomial ring $\mathbb{F}_q[t]$ has much in common with the integer ring $\mathbb{Z}$. Like $\mathbb{Z}$, it is a principal ideal domain of dimension one, and the residue fields $\mathbb{F}_q[t]/(f)$ one obtains by taking the quotient by a maximal ideal (f), where $f \in \mathbb{F}_q[t]$ is any irreducible polynomial, are finite fields $\mathbb{F}_{q^d}$, where d is the degree of f. In contrast to the situation with $\mathbb{Z}$, the residue fields of $\mathbb{F}_q[t]$ all have the same characteristic as its fraction field $\mathbb{F}_q(t)$, which plays a role analogous to $\mathbb{Q}$. Global function fields are finite extensions of $\mathbb{F}_q(t)$ (this includes $\mathbb{F}_q(t)$ itself, an extension of degree 1).

Associated to each global field k is an infinite collection of *local fields* corresponding to the completions of k with respect to its absolute values; when $k = \mathbb{Q}$, these completions are the field of real numbers $\mathbb{R}$ and the p-adic fields $\mathbb{Q}_p$ (as you will prove on Problem Set 1).

The ring $\mathbb{Z}$ is a principal ideal domain (PID), as is $\mathbb{F}_q[t]$, and in such fields every nonzero prime ideal is maximal and thus has an associated *residue field*. For both $\mathbb{Z}$ and $\mathbb{F}_q[t]$ these residue fields are finite, but the characteristics of the residue fields of $\mathbb{Z}$ are all different (and distinct from the characteristic of its fraction field), while those of $\mathbb{F}_q[t]$ are all the same.